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1.Introduction

Logistics and supply chain operations generate a large amount of historical data related to orders, shipments, customers, products, locations, and delivery performance. Analyzing this data using machine learning can help logistics organizations identify meaningful patterns, assess delivery risks, and support data-driven operational planning and decision-making.

1.1 Track

Chosen Track: Guided Data Track

Domain: Logistics and Supply Chain

Dataset: DataCo Smart Supply Chain

Business Scenario: “A logistics company wants to improve delivery reliability and reduce operational delays.”

This study aims to analyze historical supply chain data and develop a machine learning solution for predicting delivery delays. The proposed system will focus on predicting whether a shipment is likely to be Late or On Time, enabling the identification of shipments that may require greater operational attention.

The system will follow a structured machine learning workflow covering data understanding, exploratory data analysis, data quality assessment, preprocessing, feature engineering, final data preparation, model development, model comparison, evaluation and recommendation. Particular attention will be given to identifying relevant predictors, handling data quality issues, addressing class imbalance where necessary, and avoiding target and temporal data leakage to ensure that the prediction process reflects information that would realistically be available when making delivery risk decisions.

2. Business Problem and Analytical Lenses

2.1 Business Problem

Supply chain operations involve multiple activities including provisioning, production, sales, and commercial distribution. The DataCo Smart Supply Chain dataset provides historical supply-chain information that can be used to investigate operational patterns and support machine learning based decision making. The dataset contains structured supply chain data and includes information related to different product categories such as sports, clothing.

A logistics company wants to use historical shipment and order data to support operational delay analysis and strategic supply chain decision making.

The proposed solution aims to:

- Analyze historical shipment fulfillment outcomes and delivery performance metrics.
- Identify patterns in operational delays across shipping modes, order regions, and product categories.
- Compare the delivery reliability and risk profiles across different geographic distribution routes and shipping tiers.
- Predict the likelihood of late delivery for future shipments using pre – fulfillment order information.
- Provide data driven insights to support resource allocation, route optimization, and proactive customer service planning.

The problem is formulated as a binary classification problem with two possible outcomes:

- Late Delivery (1)
- On – Time / Early Delivery (0)

2.2 Primary Analytical Lens – Late Delivery Risk Prediction

This lens focuses on analyzing historical shipment fulfillment outcomes and predicting the likelihood of an unseen order experiencing a late delivery as a Binary Classification task (Late delivery vs. On Time or Early delivery).

Reasons for Selection:

- Directly addresses the main objective of the logistics and supply chain scenario.
- Provides a clear and measurable machine learning target (Late_delivery_risk). Also, if required the ability to use Delivery Status feature to verify it and document any inconsistencies found.
- Supports analysis of key factors associated with operational supply chain delays.
- Can provide actionable decision support information for the logistics organization to reduce penalties and improve delivery reliability.

2.3 Secondary Analytical Lens – Shipment Prioritization

The lens focuses on combining the predicted late delivery risk score with the business impact metrics such as order value and profit margin too establish an actionable prioritization ranking for flagged shipments.

The analysis will consider factors such as:

- Model predicted probability of late delivery.
- Order financial metrics, including total sales value (Sales) and profit per order (Benefit per order), order urgency.
- High value versus low value order risk exposure.
- Composite risk value priority scoring.
- Resource allocation tiers for operational triage and proactive customer service intervention.

Reasons for Selection:

- Directly extends the primary lens without introducing disconnected analysis, maintaining strong methodological coherence.
- Solves a real - world operational constraint where hundreds of shipments are flagged as “At Risk” logistics teams cannot intervene on all of them simultaneously.
- Ensures operational intervention focuses on shipments with the highest financial and strategic risk and urgency to the business.
- Perfectly complements the primary Late Delivery Risk Prediction lens by translating raw machine learning probability outputs into actionable business decisions.

3. Proposed Task and Output Definition

3.1 Proposed Task

The proposed task is a Supervised Binary Classification Task.

The objective is to develop a machine learning model that predicts the probability of an order experiencing a late delivery using historical order data available strictly at the time of order placement.

Input:

The feature scope is strictly restricted to fields knowable at order time:

- **Shipping and Order Context:** Shipping Mode, Type (Payment method) and scheduled fulfillment timeline (Days for shipment(scheduled)).
- **Geographic Information:** Destination Order Region, Order Country, Order City, Order State and customer location attributes.
- **Product and Order Attributes:** Category Id, Category Name, order item quantity, item prices and total sales (Sales).
- **Customer Profile Data:** Customer Segment, geographic location and historical customer context.
- **Temporal Features:** Order date/time components

3.2 Targets and Outputs

The target variable is derived from the Late_delivery_risk field where 1 = Late delivery risk identified / shipping cancelled and 0 = On Time or advance shipping.

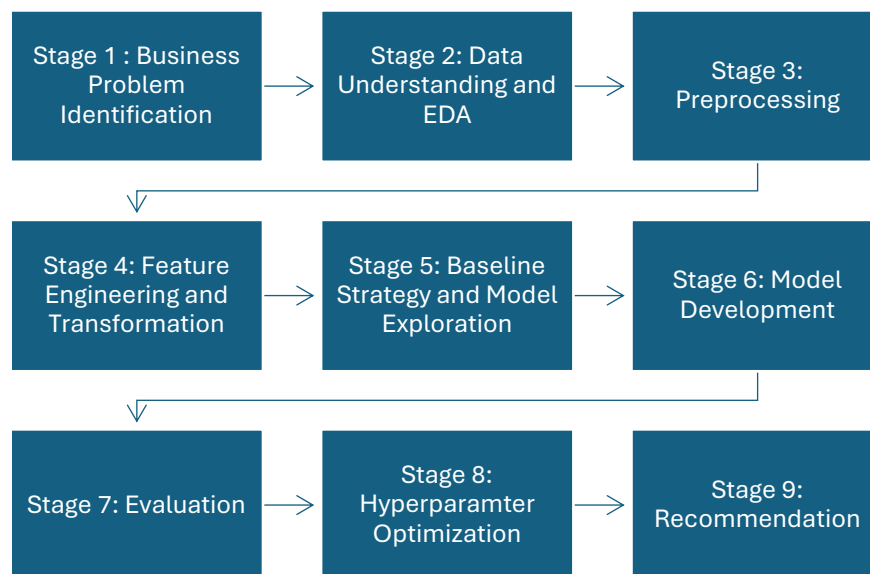
Primary Output: A predicted probability score of late delivery per order generated at order time.

Secondary Output: A priority score and rank per order which is calculated by combining the predicted risk probability with the business value metrics. This produces a ranked action list for the operations team to address high value risk orders first.

4. Proposed Workflow

The proposed system will develop a machine-learning-based late-delivery risk prediction and shipment prioritization decision-support tool. The system will learn from historical shipment and order transaction data to predict whether a new order is at risk of experiencing an operational delay. It will combine predicted delay probabilities with order profitability and urgency to rank and prioritize high-value at-risk orders, enabling logistics operations teams to proactively intervene before fulfillment delays occur.

4.1 Workflow Flow Chart



4.2 Detailed Stage Breakdown

4.2.1 Stage 1: Business Problem Identification

- The project begins by understanding the company's existing logistics and delivery process. The team focuses on identifying key operational parameters:
- Who currently monitors fulfillment delays and manages customer guarantees.
- What information is evaluated when dispatching orders across various shipping modes and geographic regions.
- What challenges and financial costs are associated with late deliveries and breaches.
- What characteristics define a high-risk shipment profile.

4.2.2 Stage 2: Data Understanding and EDA

The raw dataset will be thoroughly analyzed to understand structural features, data quality and delay patterns. The analysis will include:

- Data size, schema validation and unit of analysis aggregation from line items to unique instances.
- Variable types
- Missing values and duplicate entry audit.
- Target class distribution auditing.
- Geographic distribution of orders.
- Temporal trends across hours, days of week.

4.2.3 Stage 3: Data Preprocessing

The dataset will be cleaned and transformed to establish a robust modeling pipeline. Preprocessing steps include:

- **Unit of Analysis Aggregation:** Aggregating line items to the single order level (Order Id).
- **Missing Value & Outlier Handling:** Imputing non-systemic missing attributes and capping extreme numerical outliers in order value or profit.
- **Categorical Standardization:** Standardizing text entries across categorical variables.
- **Data Splitting:** Executing a stratified train-test and validation split prior to encoding and scaling to maintain class ratios and eliminate split leakage.

4.2.4 Stage 4: Feature Engineering & Transformation

Feature engineering will convert raw order, product, and regional data into inputs optimized for machine learning algorithms.

Order & Shipping Context Features:

- Ratio of order volume to scheduled shipping duration.
- Express shipping indicators derived from Shipping Mode.
- Payment type indicators (Type).

Geographic & Category Delay Indicators:

- Regional historical delay rate (target-encoded average delay rate per Order Region or Order Country derived strictly from training data).
- Product category risk profiles based on historical fulfillment performance (Category Name).

Temporal Features:

- Extracted order time variables: order hour, day of week, month, and weekend/holiday flags.
- Seasonality indicators to capture holiday shipping congestion.

Financial & Priority Features (Secondary Lens Support):

- Total aggregated order value (Sales).
- Aggregate order profit margin (Benefit per order).
- Composite risk-value priority metric.

4.2.5 Stage 5: Baseline Strategy & Model Exploration

Before training machine learning algorithms, a non-ML baseline strategy will be established to provide a benchmark for model comparison.

A suitable baseline for this project will be a Rule-Based Historical Frequency Classifier (e.g., predicting an order as "Late" if its Shipping Mode or Order Region historically exceeded a 50% delay rate in the training set).

Baseline performance will be evaluated on the held-out test set using the same metrics applied to ML models.

4.2.6 Stage 6: Model Development

According to the lenses selected we will mainly focus on below models but according to the features and outputs we will analyze other suitable models and select the best performing model.

4.2.6.1 Logistic Regression

Logistic Regression will serve as a linear baseline classification model. It provides high interpretability, calibrated output probabilities, and a baseline comparison for non-linear structures.

4.2.6.2 Random Forest

Random Forest will be evaluated as a non-linear tree-based ensemble model. It effectively captures non-linear interactions between shipping modes, regions, and product categories without strict parametric assumptions, while supplying global feature importance metrics.

4.2.6.3 XGBoost

XGBoost will be evaluated as the primary advanced gradient-boosted decision tree algorithm. It is well-suited for tabular supply chain data, offering handling of missing values, regularized learning to prevent overfitting, and high predictive precision. Model selection will be guided by predictive performance, probability calibration, interpretability, computational efficiency, and business applicability.

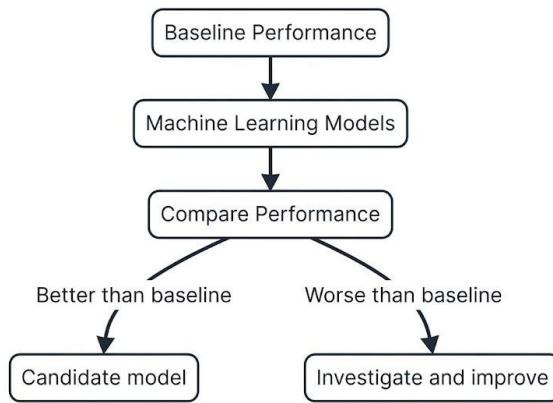


Figure 1: Model Selection

4.2.7 Stage 7: Evaluation

Models will be evaluated on a held-out test dataset. Because missing a late delivery (False Negative) result in unmanaged penalties and customer dissatisfaction, **Recall (Sensitivity)** will serve as the primary metric, supported by complementary metrics:

- **Recall:** Proportion of actual late deliveries correctly identified.
- **Precision:** Accuracy of positive late delivery flags.
- **F1 Score:** Harmonic mean balancing Precision and Recall.
- **ROC-AUC and PR-AUC:** Measurement of probability discrimination capability across thresholds.

4.2.8.A Stage 8: Hyperparameter Optimization

Hyperparameters optimization will be conducted using Optuna on candidate models. K-fold cross validation will be executed during optimization if the time and resources are feasible. Also, we will split the data as validation split according to the prevailing conditions.

The optimization objective will maximize validation Recall while tuning classification decision thresholds to optimize operational efficiency.

4.2.8.B SHAP Analysis

Calibrate prediction probabilities and perform SHAP feature attribution analysis.

4.2.9 Stage 9: Recommendation

The final phase will translate analytical outputs into business recommendations for logistics stakeholders:

- Demonstration of the superior model architecture and performance improvements over baseline benchmarks.
- Discussion of model limitations, data scope restrictions.

The system will be positioned as a human-in-the-loop decision-support tool

designed to augment supply chain oversight.

5. Core Responsibilities of Group Members

Member	Stage of the workflow	Core Responsibilities
IT24102372 Wijesinghe D H R	Business Problem Identification and Recommendation Data Understanding and EDA	Lead business problem framing canvas, business lens alignment and decision support logic. Conduct risk ethical and operational constraint analysis. Synthesize final recommendations. Perform data quality audits, missing value checks, and target distribution audits. Conduct EDA to map historical delay patterns across shipping modes, regions, product categories, and temporal periods.
IT24102364 Sanjana K D A	Data Preprocessing and Feature Engineering	Built unit of analysis aggregation pipeline, collapsing multi line items to single order id instances. Execute data cleaning, missing value imputation, outlier treatment and categorical encodings. Implement strict data leakage prevention. Extract temporal features and setup stratified train-test splits.
IT24101446 Wijekoon W M N S B	Baseline Establishment and Model Development	Establish rule-based non-ML baseline benchmarks and develop candidate ML classifiers Implement suitable models by identifying the output and

		features and analyzing the best model. Cross-validation and model training. Serialize trained model artifacts and export model pipelines for real-time application deployment.
IT24101303 Pahalawaththage P W I H	Model Evaluation and Hyperparamteter Optimization	Execute hyperparameter optimization using Optuna. Evaluate models across classification metrics (Recall, Precision, F1, ROC-AUC) and ranking metrics (Top-K High-Value Hit Rate). Calibrate prediction probabilities and perform SHAP feature attribution analysis.

6. Expected Contributions and Significance of the Study

The study is expected to contribute to supply chain and logistics analytics by transforming historical order transaction data into a structured machine learning solution for late delivery risk prediction and shipment prioritization.

1. Data-Driven Logistics Analysis

- The project will provide a systematic, empirical approach to analyzing fulfillment performance and delivery bottlenecks rather than relying on subjective operational assumptions.

2. Delay Pattern Analysis

- The engineered geographic, categorical, and order-time features will provide a structured way to evaluate delay factors across different shipping modes, distribution regions, and product categories.

3. Future Delivery Risk Prediction & Shipment Prioritization

- The final classification model will output a calibrated delay probability for unseen orders.
- By combining predicted delay probabilities with order profitability and urgency of the order, the system will generate a ranked priority action queue to optimize operational intervention.

4. Leakage – Aware Machine Learning Pipeline

- The project places strong emphasis on preventing target and temporal leakage.

The model will complement existing logistics management by providing consistent, data – driven predictions and automated prioritization to mitigate the penalties.

7. Conclusion

This study addresses the Guided Data Track – Logistics and Supply Chain business problem of using historical order and fulfillment data for late delivery analysis and strategic operational decision making.

The proposed solution focuses primarily on Late Delivery Risk Prediction, with Shipment Prioritization as the secondary analytical lens. The task is formulated as a supervised binary classification problem in which the model predicts whether an unseen order is at risk of experiencing a late delivery.

The proposed workflow covers the complete machine learning pipeline from order level data aggregation and EDA to data leakage prevention, feature engineering, multi model evaluation etc. Strict data leakage prevention and input filtering will be applied to ensure the prediction task strictly utilizes order time features.

The expected outcome is a machine learning driven decision support tool that provides supply chain planners with automated, value ranked action queues and transparent risk explainability to minimize penalties and improve delivery reliability.